

Tumorigenesis and Tumor Progression

MAPK Signaling Pathway in Melanoma

Raf phosphorylates and activates Mek, which in turn phosphorylates MAPK (i.e., Erks). This signaling cascade serves to intracellularly amplify extracellular signals mediated by growth factors. There are three functional Raf proteins in humans: A-Raf, B-Raf, and C-Raf. B-Raf has significantly higher basal kinase activity compared to A-Raf and C-Raf. Somatic mutations in *BRAF* occur at moderate to high frequencies in melanoma, as well as in colorectal, ovarian, and papillary thyroid carcinomas, indicating that activating oncogenic mutations in *BRAF* are key drivers of malignancy.

BRAF mutations are more common in melanomas arising on skin with intermittent sun exposure. Approximately 81% of melanomas on skin without chronic sun damage harbor either *BRAF* or *NRAS* mutations. In contrast, most melanomas originating from acral sites (palms, soles, subungual regions), mucosa, or chronically sun-damaged skin lack mutations in these genes. Genomic copy number variations also differ by anatomical site. Melanomas with wild-type *BRAF* or *RAS* often show amplification of *CDK4* and *CCND1* (cyclin D1), two critical G1 cell cycle regulators that are transcriptionally activated by MAPK signaling.

PTEN and Akt Pathway

Mutations in the *PTEN/MMAC1* tumor suppressor gene lead to enhanced Ras pathway signaling. PTEN is a protein/lipid phosphatase that negatively regulates the Akt pathway downstream of Ras. Loss of PTEN function results in constitutive Akt activation, promoting increased cell survival and resistance to apoptosis.

Other Genes Involved in Melanoma

Microphthalmia-Associated Transcription Factor (MITF)

Somatic amplifications of the *MITF* gene have been identified in a subset of melanomas. Overexpression of MITF, particularly in combination with activated *BRAF*, can drive melanocyte transformation. MITF is a master transcription factor regulating melanocyte differentiation, and its amplification contributes to a carcinogenic mechanism termed **lineage addiction**, where tumor cells become dependent on lineage-specific transcription factors for survival and proliferation.

Melanocortin-1 Receptor (MC1R)

The red hair/fair skin phenotype, a known risk factor for cutaneous melanoma, is strongly associated with germline mutations in the *MC1R* gene. These variants increase melanoma risk two- to fourfold in the general population and also confer increased risk for nonmelanoma skin cancers. Due to the high prevalence of *MC1R* mutations in individuals of European ancestry, the population-attributable risk for melanoma is greater than that of rarer, high-penetrance *CDKN2A* mutations.

Xeroderma Pigmentosum (XP) Genes

Xeroderma pigmentosum (XP) is a rare autosomal recessive disorder caused by mutations in at least seven genes involved in nucleotide excision repair (NER), a pathway responsible for removing UV-induced DNA photoproducts. Defects in NER lead to increased DNA mutagenesis and early onset of skin cancers. Patients with XP have a 600- to 1000-fold increased risk of developing skin malignancies, including cutaneous melanoma.

BRCA2

Carriers of germline *BRCA2* mutations have an approximately 2.58-fold increased risk of melanoma. While epidemiological data suggest a link between

breast cancer and melanoma, the biological basis for this association remains unclear and requires further investigation.

Stages of Melanomagenesis

Five stages of malignant transformation and tumor progression in melanocytes have been proposed, based on clinical, histopathological, immunological, cytogenetic, and in vitro evidence:

1. **Benign melanocytic nevi**
2. **Atypical (dysplastic) nevi**
3. **Primary malignant melanoma, radial growth phase**
4. **Primary malignant melanoma, vertical growth phase**
5. **Metastatic malignant melanoma**

Each stage is thought to represent clonal evolution, where successive genetic alterations confer growth advantages, leading to clonal expansion and progression to more aggressive phenotypes.